Cancer mortality in poultry slaughtering/processing plant workers belonging to a union pension fund.

BACKGROUND: The role of zoonotic biological agents in human cancer occurrence has been little studied. Humans are commonly exposed to viruses that naturally infect and cause cancer in food animals such as poultry that constitute part of the biological environment. It is not known if these viruses cause cancer in humans. OBJECTIVE: To study cancer mortality in the largest cohort to date, of 20,132 workers in poultry slaughtering and processing plants, a group with the highest human exposures to these viruses. METHODS: Mortality in poultry workers was compared with that in the US general population through the estimation of standardized mortality ratios. RESULTS: Significantly increased risks were observed in the cohort as a whole or in subgroups, for several cancer sites, viz: cancers of the buccal cavity and pharynx; pancreas; trachea/bronchus/lung; brain; cervix; lymphoid leukemia; monocytic leukemia; and tumors of the hemopoietic and lymphatic systems. Elevated SMRs that were not statistically significant were observed for cancers of the liver, nasopharynx, myelofibrosis, and myeloma. New sites observed to be significantly in excess in this study were cancers of the cervix and penis. CONCLUSION: This large study provides evidence that a human group with high exposure to poultry oncogenic viruses has increased risk of dying from several cancers. Other occupational carcinogenic exposures could be of importance in explaining some of the findings, such as fumes from wrapping machines. These findings may have implications for public health amongst persons in the general population who may also be exposed to these viruses. What is needed now are epidemiologic studies that can demonstrate whether the excess of specific cancers can be attributed to specific occupational exposures while adequately controlling for other potential occupational and non-occupational carcinogenic exposures.